Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [13AE65/12AE65]

SIXTH SEMESTER B.Tech/B.E UNIVERSITY EXAMINATION, JUNE – 2016

FRACTURE MECHANICS

Time: 3 Hours **Maximum Marks:70**

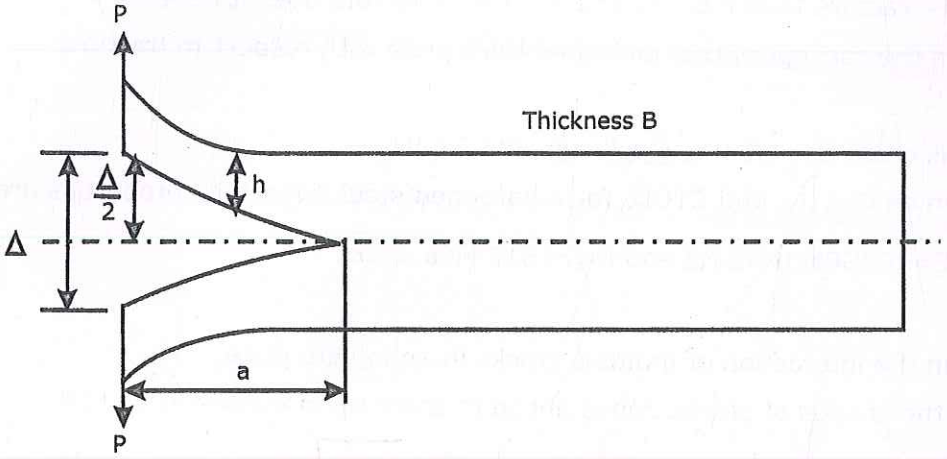
Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. a. With a neat sketch explain the different types of modes of fracture. **07**

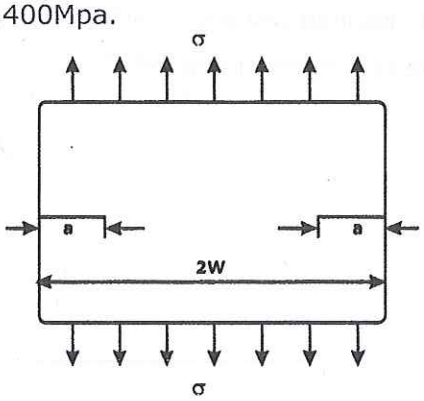
b. Bring out the difference between the types of brittle fracture. **07**

2. Derive strain energy release rate 'G' for a double cantilever bean show in figure below



Determine the critical energy release rate of the above DCB specimen if thickness is 30mm, depth of each cantilever is half of the thickness of the specimen and crack length is twice the value of thickness of the DCB specimen. It is made up of hardened steel with the modulus (207GPa) and the crack is about to propagate at 1500N, pulling load. **14**

3. a. Determine the mode – I Stress Intensity Factor (K_1) for a double edged cracked plate under uniform tension shown below. Also calculate the mode – I strain energy release rate if the material is made up of aluminum whose modulus is 72,400Mpa. **07**



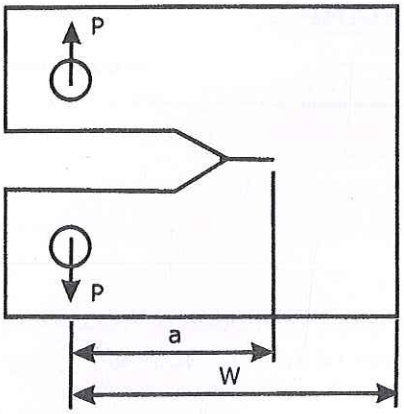
σ = 120MPa

2W = 100mm

A = 10mm

$f(\alpha) = 1.12-0.2\alpha -1.20\alpha^2+1.93\alpha^3$

b. Calculate the thickness and mode – I strain energy release rate of a compact test specimen shown below whose geometric correction factor is 0.97 and elastic modulus is 200GPa. **07**



P = 10,000N

a = 15mm

W = 100mm

K_I = 450 MPa √mm

4. **a.** What is fracture toughness? Explain on what factors does it depend. **07**

b. Explain the concept of thin plate and thick plate with respect to fracture. **07**

5. **a.** What is crack-tip-opening displacement? Explain. **04**

b. Determine G_{1C}, J_{1C} and $CTOD_C$ for a hardened steel if material properties are J_{ys}=700 MPa, E=207GPa. υ =0.33 and K_{1c}=1517 MPa $\sqrt{mm}$ **10**

6. **a.** Explain the interaction of multiple cracks in an infinite plate. **07**

b. Draw the shapes of plastic zones ahead of crack tip in mode-I,II and III **07**

7. **a.** Explain the effect of an overload pulse on the constant amplitude fatigue load with respect to crack propagation. **07**

b. A pressure vessel is made of aluminum alloy and operates in two stages at 20 cycles per minute. It operates in the first stage 60% of the time at a stress change of 1500 psi; the cycles to failure in this stage are 1E6 cycles. In stage 2 (40% of the time) a stress change of 2400Psi is exerted on the vessel and cycles to failure in stage 2 are 215E3. Find the expected life in hours for this hydraulic unit. **07**

8. **a.** What is Non-destructive testing (NDT)? Explain the steps involved in NDT. **07**

b. Explain the ultra sonics non destructive testing with neat sketch. **07**